

Normal approximation of the Poisson distribution

III Recall: $X \sim \text{Poi}(\lambda)$, then

$$P_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

III Let $X_i \sim \text{Poi}(\lambda_i)$ which are independent

then $S = X_1 + \dots + X_n \sim \text{Poi}\left(\sum_{i=1}^n \lambda_i\right)$.

III A motivation example: $X \sim \text{Poi}(100)$,

$$P(X > 200) = \sum_{x=201}^{\infty} \frac{(100)^x e^{-100}}{x!}$$

$$= 1 - \sum_{x=0}^{200} \frac{(100)^x e^{-100}}{x!}$$

which is
computed numerically.

Set-up for the normal approximation:

• Use CLT : $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \stackrel{\sim}{=} Z$

• Write $X = X_1 + \dots + X_n$, where

$X_i \sim \text{Poi}\left(\frac{\lambda}{n}\right)$, so $X \sim \text{Poi}\left(\sum_{i=1}^n \frac{\lambda}{n}\right)$. ↑ λ

• $\frac{\frac{1}{n} \cdot X - \mu}{\sigma/\sqrt{n}} = \frac{\cancel{\frac{1}{n}} X - \cancel{\frac{\lambda}{n}}}{\sqrt{\lambda}/\cancel{n}} \stackrel{\sim}{=} Z$

• $\mu = E(X_i) = \frac{\lambda}{n}$

• $\sigma^2 = \text{Var}(X_i) = \frac{\lambda}{n}$

$$\therefore X \stackrel{\circ}{\approx} \sqrt{\lambda} \cdot Z + \lambda \sim N(\lambda, \lambda)$$

Ex. $X \sim \text{Poi}(100)$, we have

$$P(X > 200) \approx P(Y > 200), Y \sim N(100, 100)$$

$$= P\left(\frac{Y - 100}{\sqrt{100}} > \frac{200 - 100}{\sqrt{100}}\right)$$

$$= P(Z > 10)$$

$$= 1 - \Phi(10)$$

$$\approx 0.$$